

H SRIPADA RAO

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Research Summary

M.Tech CSE student researching explainable AI for cloud anomaly detection, with a working system that pairs unsupervised detection (Isolation Forest) with SHAP-based root-cause explanations on real AWS CloudWatch metrics. Research interests: explainable machine learning, anomaly detection, and reliable cloud systems. Hands-on experience with AWS, Terraform, and Python.

Education

Dr. Ambedkar Institute of Technology, Bengaluru **Expected 2027**
M.Tech in Computer Science and Engineering — CGPA: 7.94/10.0

Vivekananda Institute of Technology **May 2025**
BE in Computer Science and Engineering with Honours — Final CGPA: 8.2/10.0

VKJ PU College **2021**
Pre-University (PCMB) — 80%

Research Project

ExplainAlert: Explainable Anomaly Detection for Cloud Servers **June 2026**

M.Tech Research Project, guided by Dr. Srinivas A H — AWS EC2, CloudWatch, Python, Isolation Forest, SHAP

- **Problem:** Cloud monitoring tools flag anomalies but rarely explain them, so non-expert operators struggle to find the cause.
- **Detection:** Built an anomaly detection system on AWS EC2 using Isolation Forest, reaching 97.9% accuracy and 100% recall on 239 labeled CloudWatch samples (CPU, network, disk) collected over 20 hours, with anomalies injected using Linux stress tools.
- **Explainability:** Integrated SHAP to give per-anomaly root-cause explanations with the percentage contribution of each metric.
- **Alerting:** Built an automated alert pipeline (AWS SES, Telegram Bot API) that sends real-time notifications with remediation steps, running entirely within the AWS Free Tier.
- **Comparison:** Against native CloudWatch anomaly detection, adds root-cause explanation, contribution analysis, and corrective-action suggestions. Extensible to other time-series monitoring domains such as disaster warning.

Other Projects

Multi-Cloud Infrastructure Automation (IaC) **2026**

Terraform, Azure Resource Manager, AWS App Runner

- Designed decoupled infrastructure on Azure and AWS using Terraform for resource provisioning.
- Structured Azure resource groups and environment stacks for departmental lab allocations.
- Deployed image-based application workloads on AWS App Runner.

High-Availability Web Server and VM Lifecycle Infrastructure **2025**

AWS EC2, Auto Scaling, Security Groups, Linux

- Created custom EC2 Linux AMIs that automate web server bootstrap.
- Isolated network layers using strict ingress/egress Security Group rules.
- Configured Auto Scaling policies driven by performance metrics to improve resilience.

Skills

ML & Explainability:	Isolation Forest, SHAP, scikit-learn, Pandas, NumPy
Cloud:	AWS (EC2, CloudWatch, SES, App Runner), Microsoft Azure
DevOps & IaC:	Terraform, Git/GitHub, Linux (Ubuntu/Fedora)
Languages:	Python, C, Bash, SQL, JavaScript
Tools:	Streamlit, boto3, VS Code, Linux CLI

Coursework

Machine Learning, Cloud Architecture, DevOps Practices, Data Structures & Algorithms, Database Management Systems, Linux System Administration

Certifications (NPTEL)

- Edge Computing (Mar 2026)
- Cloud Computing (May 2024)
- The Joy of Computing using Python (Oct 2023)
- Problem Solving through Programming in C (Oct 2023)